# Sequences, Series & Complex Numbers

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**ANSWER KEY — COMPLETE WORKED SOLUTIONS**

## Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

**1** **[Foundational]** • **Target: Arithmetic sequences**

Find the 12th term of the arithmetic sequence $5,\ 9,\ 13,\ \ldots$

$a_1 = 5$, $d = 4$. $a_{12} = 5 + (11)(4) = 49$.

**2** **[Foundational]** • **Target: Arithmetic sequences**

The 4th term of an arithmetic sequence is 18, and the 9th term is 38. Find the first term and the common difference.

$a_1 + 3d = 18$ and $a_1 + 8d = 38$. Subtracting: $5d = 20 \Rightarrow d = 4$, so $a_1 = 18 - 12 = 6$.

**3** **[Foundational]** • **Target: Arithmetic sequences**

Find the sum of the first 20 terms of the arithmetic sequence $7,\ 11,\ 15,\ \ldots$

$a_1 = 7$, $d = 4$, $a_{20} = 7 + (19)(4) = 83$. $S_{20} = \frac{20}{2}(7 + 83) = 10(90) = 900$.

**4** **[Foundational]** • **Target: Arithmetic sequences**

Find the number of terms in the arithmetic sequence $6,\ 10,\ 14,\ \ldots,\ 150$.

$a_1 = 6$, $d = 4$. $6 + 4(n - 1) = 150 \Rightarrow 4(n - 1) = 144 \Rightarrow n - 1 = 36 \Rightarrow n = 37$.

**5** **[Foundational]** • **Target: Geometric sequences**

Find the 7th term of the geometric sequence $3,\ 6,\ 12,\ \ldots$

$a_1 = 3$, $r = 2$. $a_7 = 3(2^6) = 3(64) = 192$.

**6** **[Foundational]** • **Target: Geometric sequences**

In a geometric sequence with all positive terms, the 3rd term is 20 and the 5th term is 80. Find $r$ and $a_1$.

$a_1 r^2 = 20$ and $a_1 r^4 = 80$. Dividing: $r^2 = 4$, and since all terms are positive, $r = 2$. Then $a_1 = \frac{20}{4} = 5$.

**7** **[Foundational]** • **Target: Geometric sequences**

Given that $x - 1$, $x + 1$, and $3x - 1$ are consecutive terms of a geometric sequence, find the possible value(s) of $x$.

$(x + 1)^2 = (x - 1)(3x - 1) \Rightarrow x^2 + 2x + 1 = 3x^2 - 4x + 1 \Rightarrow 0 = 2x^2 - 6x = 2x(x - 3)$. So $x = 0$ or $x = 3$. If $x = 0$, the terms are $-1,\ 1,\ -1$ (ratio $-1$, valid). If $x = 3$, the terms are $2,\ 4,\ 8$ (ratio $2$, valid).

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## Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

**8** **[Intermediate]** • **Target: Geometric sequences**

Find the sum of the first 6 terms of the geometric sequence $5, 15, 45, \ldots$

$a_1 = 5, r = 3. S_6 = \frac{5(3^6 - 1)}{3 - 1} = \frac{5(728)}{2} = 1820.$

**9** **[Intermediate]** • **Target: Series and sigma notation**

Write the series $3 + 7 + 11 + \cdots + 39$ in sigma notation, then evaluate it.

The general term is $a_k = 4k - 1$ for $k = 1, \ldots, 10$ (since $4(10) - 1 = 39$). $\sum_{k=1}^{10} (4k - 1) = 210.$

**10** **[Intermediate]** • **Target: Geometric series**

Find the sum to infinity of $16 - 8 + 4 - 2 + \cdots$

$a_1 = 16, r = -\frac{1}{2}. S_\infty = \frac{16}{1 - \left(-\frac{1}{2}\right)} = \frac{16}{3/2} = \frac{32}{3}.$

**11** **[Intermediate]** • **Target: Geometric series**

Express the repeating decimal $0.4\overline{5}$ (that is, $0.45555\ldots$) as a geometric series and find its exact fractional value.

$0.4\overline{5} = 0.4 + (0.05 + 0.005 + 0.0005 + \cdots)$. The bracketed part is geometric with $a_1 = 0.05$, $r = 0.1$, summing to $\frac{0.05}{0.9} = \frac{1}{18}$. Total: $\frac{2}{5} + \frac{1}{18} = \frac{36}{90} + \frac{5}{90} = \frac{41}{90}$.

**12** **[Intermediate]** • **Target: Complex number arithmetic**

Simplify $i^{23}$, using $i^2 = -1$.

Powers of $i$ cycle with period 4: $i^1 = i$, $i^2 = -1$, $i^3 = -i$, $i^4 = 1$. Since $23 = 4(5) + 3$, $i^{23} = i^3 = -i$.

**13** **[Intermediate]** • **Target: Complex conjugates and division**

Find the conjugate of $z = 6 + 7i$, and compute $z\bar{z}$.

$\bar{z} = 6 - 7i. z\bar{z} = 6^2 + 7^2 = 36 + 49 = 85.$

**14** **[Intermediate]** • **Target: Complex conjugates and division**

Simplify $\frac{4 + 3i}{1 - 2i}$ by multiplying by the conjugate of the denominator.

$\frac{4+3i}{1-2i} \cdot \frac{1+2i}{1+2i} = \frac{(4+3i)(1+2i)}{1^2 + 2^2} = \frac{4 + 8i + 3i + 6i^2}{5} = \frac{-2 + 11i}{5} = -\frac{2}{5} + \frac{11}{5}i.$

## Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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**15** [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Solve $x^2 + 4x + 13 = 0$, giving your answer in the form $a + bi$.

By the quadratic formula: $x = \frac{-4 \pm \sqrt{16 - 52}}{2} = \frac{-4 \pm \sqrt{-36}}{2} = \frac{-4 \pm 6i}{2} = -2 \pm 3i$.

**16** [Challenging] • Target: Quadratic equations with complex roots • Cross-Lesson: 5.01 Trigonometry

One root of a quadratic equation with real coefficients is $2 - 5i$. Find the other root, and write the equation in standard form.

Since the coefficients are real, the other root is $2 + 5i$. Sum of roots $= 4$; product $= (2 - 5i)(2 + 5i) = 4 + 25 = 29$. Equation: $x^2 - 4x + 29 = 0$.

**17** [Challenging] • Target: A well-designed word problem: a bouncing ball • Cross-Lesson: 6.01 Circles & Coordinate Geometry

This question has three parts. A ball is dropped from a height of 20 meters. Each time it bounces, it rises to 60% of its previous height. (a) Find the height of the ball's rebound after its 4th bounce. (b) Write an expression for the total vertical distance traveled by the ball as it bounces infinitely many times, and evaluate it. (c) After which bounce does the rebound height first fall below 1 meter?

(a) The rebound height after the $n$th bounce is $h_n = 20(0.6)^n$. $h_4 = 20(0.6)^4 = 20(0.1296) = 2.592$ meters. (b) The ball falls 20 m initially, then for each bounce it rises and falls the same rebound height. Total distance $= 20 + 2\sum_{n=1}^{\infty} 20(0.6)^n = 20 + 2\left(\frac{12}{1-0.6}\right) = 20 + 2(30) = 80$ meters. (c) We need $20(0.6)^n < 1 \Rightarrow (0.6)^n < 0.05$. Checking: $(0.6)^5 = 0.0778$ gives $h_5 = 1.555$ m (still above 1 m), while $(0.6)^6 = 0.0467$ gives $h_6 = 0.933$ m (below 1 m). The rebound height first falls below.

**18** [Challenging] • Target: A well-designed word problem: a bouncing ball • Cross-Lesson: 7.01 Solid Geometry & Similarity

Let $z = \frac{5 + 2i}{2 - 3i} + i^{43}$. (a) Express $z$ in standard form $a + bi$. (b) Compute the modulus $|z|$.

(a) Simplify the quotient and power of $i$:First evaluate $\frac{5 + 2i}{2 - 3i}$: multiply numerator and denominator by the conjugate $2 + 3i$: $\frac{(5 + 2i)(2 + 3i)}{(2 - 3i)(2 + 3i)} = \frac{10 + 15i + 4i + 6i^2}{4 - 9i^2} = \frac{10 - 6 + 19i}{4 + 9} = \frac{4 + 19i}{13}$. Now evaluate $i^{43}$: $43 = 4(10) + 3$, remainder 3, so $i^{43} = i^3 = -i = -\frac{13i}{13}$.Combine terms: $z = \frac{4 + 19i}{13} - i = \frac{4 + 19i - 13i}{13} = \frac{4 + 6i}{13} = \mathbf{\frac{4}{13} + \frac{6}{13}i}$. (b) Modulus $|z|$.

**19** [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 8.01 Statistics & Probability

The sum of the first $n$ terms of an arithmetic sequence is given by $S_n = 2n^2 + 3n$. Find the 10th term $a_{10}$.

Step 1: Use relationship between $S_n$ and $a_n$: $a_n = S_n - S_{n-1}$ Step 2: Method 1 (Direct evaluation): $S_{10} = 2(10^2) + 3(10) = 2(100) + 30 = 230$ $S_9 = 2(9^2) + 3(9) = 2(81) + 27 = 162 + 27 = 189$ $a_{10} = S_{10} - S_9 = 230 - 189 = \mathbf{41}$. Method 2 (General $n$th term formula): $a_n = (2n^2 + 3n) - [2(n-1)^2 + 3(n-1)] = (2n^2 + 3n) - (2n^2 - 4n + 2 + 3n - 3) = 4n + 1$ For $n = 10$: $a_{10} = 4(10) + 1 = \mathbf{41}$.

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**20** [Challenging] • Target: Geometric sequences • Cross-Lesson: 1.01 Advanced Algebra

An infinite geometric series has first term $a_1 = 18$ and sum to infinity $S_\infty = 24$. (a) Find the common ratio $r$. (b) Calculate the third term $a_3$.

(a) Find common ratio $r$: $S_\infty = \frac{a_1}{1-r} \Rightarrow 24 = \frac{18}{1-r} \Rightarrow 1 - r = \frac{18}{24} = \frac{3}{4} \Rightarrow r = 1 - \frac{3}{4} = \mathbf{\frac{1}{4}}$. (b) Calculate 3rd term $a_3$: $a_3 = a_1 \cdot r^2 = 18 \cdot \left(\frac{1}{4}\right)^2 = 18 \cdot \frac{1}{16} = \mathbf{\frac{9}{8} = 1.125}$.

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## Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

**21** [Foundational] • Target: Arithmetic sequences

Find the 3rd term of the geometric sequence $2, 6, 18, \ldots$

**a.** 16

**b.** $-18$

**c.** 18 ✓

**d.** 20

$a_1 = 2, r = 3$. $a_3 = 2(3^2) = 2(9) = 18$.

**22** [Foundational] • Target: Geometric sequences

The first term of a geometric sequence is $a$. The common ratio is $r$. The sum of the first 3 terms is 21, and the sum of the first 6 terms is 189. If $r > 0$, find the value of $a$ and $r$.

**a.** $a = 3, r = 2$ ✓

**b.** $a = 7, r = 1/2$

**c.** $a = 3, r = \sqrt{2}$

**d.** $a = 21, r = 1$

Let the geometric sequence be denoted by $a_n$, where $a_1 = a$ and the common ratio is $r$. The sum of the first $n$ terms of a geometric sequence is given by $S_n = \frac{a(r^n - 1)}{r - 1}$ (for $req1$). We are given that $S_3 = 21$ and $S_6 = 189$. So, we have the equations: 1) $\frac{a(r^3 - 1)}{r - 1} = 21$ 2) $\frac{a(r^6 - 1)}{r - 1} = 189$ We can rewrite $r^6 - 1$ using the difference of squares factorization: $r^6 - 1 = (r^3)^2 - 1^2 = (r^3 - 1)(r^3 + 1)$. Substitute this into the second equation.

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**23** [Foundational] • Target: Series and sigma notation

An arithmetic sequence has its $n$-th term given by $a_n = 3n + k$, where $k$ is a constant. If the sum of the first 5 terms of this sequence is 65, find the value of $k$.

**a.** 1

**b.** **2 ✓**

**c.** 3

**d.** 4

The sum of the first $n$ terms of an arithmetic sequence is given by $S_n = \frac{n}{2}(2a_1 + (n-1)d)$ or $S_n = \frac{n}{2}(a_1 + a_n)$. In this case, $a_n = 3n + k$. So, $a_1 = 3(1) + k = 3 + k$. The common difference is $d = a_{n+1} - a_n = (3(n+1) + k) - (3n + k) = 3n + 3 + k - 3n - k = 3$. The sum of the first 5 terms is $S_5 = 65$. Using the formula $S_n = \frac{n}{2}(a_1 + a_n)$, we have $S_5 = \frac{5}{2}(a_1 + a_5)$. We need $a_5 = 3(5) + k = 15 + k$. Therefore, $65 = \frac{5}{2}((3 + k) + (15 + k))$. Multiply both sides b.

**24** [Foundational] • Target: Geometric series

Consider the arithmetic sequence where the first term is $a_1 = 5$ and the common difference is $d = 3$. What is the fifth term of this sequence?

**a.** **17 ✓**

**b.** 14

**c.** 20

**d.** 23

The formula for the $n$-th term of an arithmetic sequence is given by $a_n = a_1 + (n-1)d$, where $a_1$ is the first term and $d$ is the common difference. In this case, $a_1 = 5$ and $d = 3$. We want to find the fifth term, so $n = 5$. Substituting these values into the formula: $a_5 = 5 + (5-1) \times 3$ $a_5 = 5 + (4) \times 3$ $a_5 = 5 + 12$ $a_5 = 17$

**25** [Foundational] • Target: Complex number arithmetic

A geometric sequence has its first term $a_1 = 3$ and a common ratio $r = 2$. What is the third term, $a_3$, of this sequence?

**a.** 6

**b.** 9

**c.** **12 ✓**

**d.** 15

The formula for the $n$-th term of a geometric sequence is $a_n = a_1 \cdot r^{n-1}$. Given $a_1 = 3$ and $r = 2$, we want to find the third term, $a_3$. Substituting $n = 3$ into the formula: $a_3 = a_1 \cdot r^{3-1} = a_1 \cdot r^2$ $a_3 = 3 \cdot 2^2$ $a_3 = 3 \cdot 4$ $a_3 = 12$. Therefore, the third term is 12.

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**26** [Foundational] • Target: Complex conjugates and division

An arithmetic sequence has $a_3 = 0$ and $a_5 = -4$. Find $S_{12}$, the sum of the first 12 terms.

**a.** -92

**b.** **-84** ✓

**c.** -85

**d.** -79

$d = \frac{-4-0}{2} = -2$, $a_1 = 4$. $a_{12} = 4 + (12-1)(-2) = -18$. $S_{12} = \frac{12(4-18)}{2} = -84$.

**27** [Foundational] • Target: Modulus of complex numbers

In an arithmetic sequence, $a_4 = -1$ and $a_8 = -9$. Find the common difference $d$.

**a.** **-2** ✓

**b.** -3

**c.** -6

**d.** 6

$d = \frac{a_8 - a_4}{8-4} = \frac{-9+1}{4} = -2$.

**28** [Foundational] • Target: Quadratic equations with complex roots

Find the sum to infinity of the geometric series with $a_1 = 4$ and $r = 2/3$.

**a.** 6

**b.** **12** ✓

**c.** 19

**d.** 15

$S_\infty = \frac{a_1}{1-r} = \frac{4}{1-(2/3)} = 12$.

## Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

**29** [Intermediate] • Target: A well-designed word problem: a bouncing ball

An arithmetic sequence has first term $a_1 = -1$ and common difference $d = 4$. Find $a_6$.

**a.** 21

**b.** 15

**c.** 26

**d.** **19** ✓

$a_6 = a_1 + (n-1)d = -1 + (6-1)(4) = 19$.

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**30** [Intermediate] • Target: Arithmetic sequences

A geometric sequence has first term $a_1 = 2$ and common ratio $r = 1/2$. Find $a_5$.

**a.** -3/8

**b.** **1/8** ✓

**c.** 17/8

**d.** 3/8

$a_5 = a_1 r^{n-1} = 2(1/2)^4 = 1/8$.

**31** [Intermediate] • Target: Geometric sequences

A sequence is defined by the recursive relation $a_{n+1} = 2a_n$ with the first term $a_1 = 5$. What is the third term of this sequence?

**a.** 10

**b.** **20** ✓

**c.** 15

**d.** 25

The sequence is defined by $a_1 = 5$ and $a_{n+1} = 2a_n$. We need to find the third term, $a_3$. First, find the second term, $a_2$, using the recursive relation with $n = 1$: $a_2 = 2a_1$ $a_2 = 2(5)$ $a_2 = 10$ Next, find the third term, $a_3$, using the recursive relation with $n = 2$: $a_3 = 2a_2$ $a_3 = 2(10)$ $a_3 = 20$ Therefore, the third term of the sequence is 20.

**32** [Intermediate] • Target: Series and sigma notation

Consider the quadratic equation $z^2 - (3 + 2i)z + (1 + 3i) = 0$. Let the roots of this equation be $\alpha$ and $\beta$. If the complex number $w$ is defined such that $w = \frac{\alpha^2 + \beta^2}{2}$, find the modulus of $w$, $|w|$.

**a.** $\sqrt{5}$ ✓

**b.** $\frac{\sqrt{5}}{2}$

**c.** $\sqrt{29}$

**d.** $\frac{\sqrt{29}}{2}$

The given quadratic equation is $z^2 - (3 + 2i)z + (1 + 3i) = 0$. By Vieta's formulas, the sum of the roots is $\alpha + \beta = 3 + 2i$ and the product of the roots is $\alpha\beta = 1 + 3i$. We need to find $w = \frac{\alpha^2 + \beta^2}{2}$. We can express $\alpha^2 + \beta^2$ in terms of the sum and product of the roots: $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$. Substituting the values from Vieta's formulas: $\alpha^2 + \beta^2 = (3 + 2i)^2 - 2(1 + 3i)$ $\alpha^2 + \beta^2 = (9 + 12i + 4i^2) - (2 + 6i)$.

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**33** [Intermediate] • Target: Geometric series

Let $(a_n)_{n \geq 1}$ be an arithmetic sequence of real numbers with first term $a_1 > 0$ and non-zero common difference $d$. The terms $a_1$, $a_3$, and $a_4$ form the first three terms of a convergent geometric sequence $(g_k)_{k \geq 1}$, such that $g_1 = a_1$, $g_2 = a_3$, and $g_3 = a_4$. Let $S_\infty = \sum_{k=1}^{\infty} g_k$ denote the sum to infinity of the geometric sequence, and let $S_n = \sum_{j=1}^{n} a_j$ denote the $n$-th partial sum of the arithmetic sequence. If $M = \max_{n \geq 1} S_n$ is the maximum value attained by the partial sums of the arithmetic sequence, what is the value of $\frac{M}{S_\infty}$?

**a.** $\frac{5}{4}$ ✓

**b.** $\frac{81}{64}$

**c.** $\frac{9}{8}$

**d.** $\frac{5}{8}$

1. **Determine the relation between $d$ and $a_1$:** The terms of the arithmetic sequence are $a_n = a_1 + (n-1)d$. Therefore: $g_1 = a_1, \quad g_2 = a_3 = a_1 + 2d, \quad g_3 = a_4 = a_1 + 3d$ Since $g_1, g_2, g_3$ form a geometric progression, the geometric mean property requires: $g_2^2 = g_1 g_3 \Rightarrow (a_1 + 2d)^2 = a_1(a_1 + 3d)$ Expanding both sides: $a_1^2 + 4a_1 d + 4d^2 = a_1^2 + 3a_1 d \; a_1 d + 4d^2 = 0 \Rightarrow d(a_1 + 4d) = 0$ Since $de0$, we have $a_1 + 4d = 0$, which gives.

**34** [Intermediate] • Target: Complex number arithmetic

Consider the quadratic equation $z^2 - (3+i)z + (2+2i) = 0$, where $z$ is a complex number. Let the roots of this equation be $z_1$ and $z_2$. Determine the value of $|z_1 - z_2|^2$.

**a.** 1

**b.** 2

**c.** **4** ✓

**d.** 5

Let the quadratic equation be $az^2 + bz + c = 0$. The roots are $z_1$ and $z_2$. We know that $z_1 + z_2 = -\frac{b}{a}$ and $z_1 z_2 = \frac{c}{a}$. The discriminant of the quadratic equation is $\Delta = b^2 - 4ac$. The roots can be expressed as $z_1 = \frac{-b + \sqrt{\Delta}}{2a}$ and $z_2 = \frac{-b - \sqrt{\Delta}}{2a}$. Therefore, $z_1 - z_2 = \frac{\sqrt{\Delta}}{a}$. Consequently, $|z_1 - z_2|^2 = \left|\frac{\sqrt{\Delta}}{a}\right|^2 = \frac{|\Delta|}{|a|^2}$. In our given equation.

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**35** [Intermediate] • Target: Complex conjugates and division

Let $z_1$ and $z_2$ be the roots of the quadratic equation $x^2 - 4x + 5 = 0$. Consider a complex number $w$ such that $w = z_1^2 + z_2^2$. If $w$ is expressed in the form $a + bi$, where $a$ and $b$ are real numbers, find the value of $a - b$.

**a.** **6 ✓**

**b.** 8

**c.** 10

**d.** 12

The quadratic equation is $x^2 - 4x + 5 = 0$. By Vieta's formulas, the sum of the roots is $z_1 + z_2 = -(-4)/1 = 4$ and the product of the roots is $z_1 z_2 = 5/1 = 5$. We are interested in $w = z_1^2 + z_2^2$. We can rewrite this as $w = (z_1 + z_2)^2 - 2z_1 z_2$. Substituting the values from Vieta's formulas: $w = (4)^2 - 2(5)$ $w = 16 - 10$ $w = 6$. So, $w = 6$. When expressed in the form $a + bi$, we have $a = 6$ and $b = 0$. The question asks for $a - b$. $a - b = 6 - 0 = 6$.

**36** [Intermediate] • Target: Modulus of complex numbers

Consider the arithmetic sequence shown on the number line. What is the common difference of this sequence?

**a.** **3 ✓**

**b.** -3

**c.** 12

**d.** 6

The common difference of an arithmetic sequence is the constant value added to each term to get the next term. Observing the number line, we can see the sequence starts at 0 and progresses to 3, then to 6, and finally to 12. The difference between consecutive terms is $3 - 0 = 3$, $6 - 3 = 3$, and $12 - 6 = 6$. However, the diagram shows markings at 0, 3, 6, and 9. Let's re-examine the number line to ensure accuracy. The marks are at 0, 3, 6, and 9. The difference between consecutive terms is $3 - 0 = 3$.

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**37** [Intermediate] • Target: Quadratic equations with complex roots

Let $m$ be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots $z_1$ and $z_2$. In the complex plane, the origin $O(0)$ and the points representing $z_1$ and $z_2$ form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

**a.** $15\sqrt{3}$ ✓

**b.** $12\sqrt{3}$

**c.** $\frac{15\sqrt{3}}{2}$

**d.** $9\sqrt{3}$

We solve the problem by synthesizing Vieta's formulas with the geometric properties of complex numbers. **Step 1: Condition for an Equilateral Triangle with the Origin** In the complex plane, three points $0$, $z_1$, and $z_2$ form the vertices of an equilateral triangle if and only if one non-zero vertex is obtained by rotating the other about the origin by $\pm\frac{\pi}{3}$: $\frac{z_2}{z_1} = e^{\pm i\pi/3} = \frac{1 \pm i\sqrt{3}}{2}$ This is equivalent to the quadratic in $\frac{z_2}{z_1}$.

## Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

**38** [Challenging] • Target: A well-designed word problem: a bouncing ball • Cross-Lesson: 7.01 Solid Geometry & Similarity

In a geometric sequence of positive real numbers, the sum of the first two terms is $18$, and the sum of the first four terms is $90$. What is the fifth term of the sequence?

**a.** 48

**b.** 72

**c.** 96 ✓

**d.** 192

Let $a_1$ be the first term and $r$ be the common ratio of the geometric sequence. Since all terms are positive real numbers, $a_1 > 0$ and $r > 0$. The sum of the first two terms is: $S_2 = a_1 + a_1 r = a_1(1 + r) = 18$ The sum of the first four terms is: $S_4 = a_1 + a_1 r + a_1 r^2 + a_1 r^3 = a_1(1 + r)(1 + r^2) = 90$ Substituting $a_1(1 + r) = 18$ into the equation for $S_4$: $18(1 + r^2) = 90$ $1 + r^2 = 5 \Rightarrow r^2 = 4$ Since $r > 0$, we have $r = 2$. Now substitute $r = 2$ back to solve for $a_1$.

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**39** [Challenging] • Target: Arithmetic sequences • Cross-Lesson: 8.01 Statistics & Probability

Consider the quadratic equation $2x^2 - (k-1)x + (k+1) = 0$ where $k$ is a real constant. The equation has two non-real complex conjugate roots. If the imaginary parts of the roots have a magnitude of $\frac{\sqrt{7}}{2}$, what is the greatest possible value of $k$?

**a.** **7** ✓

**b.** 3

**c.** 10

**d.** 5

To find the parameter $k$, apply the quadratic formula to $ax^2 + bx + c = 0$: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ Identify the coefficients from the given equation: $a = 2, \quad b = -(k-1), \quad c = k+1$ Compute the discriminant $\Delta$: $\Delta = [-(k-1)]^2 - 4(2)(k+1) = (k^2 - 2k + 1) - (8k + 8) = k^2 - 10k - 7$ Since the equation has non-real complex roots, $\Delta 0$, confirming the roots are non-real. The greatest possible value of $k$ is $7$. - Misconception 3 arises from selecting the smalle.

**40** [Challenging] • Target: Geometric sequences • Cross-Lesson: 1.01 Advanced Algebra

Consider the quadratic function $f(x) = x^2 - 6x + 8$. What are the roots of this function?

**a.** $x = 2$ **and** $x = 4$ ✓

**b.** $x = -2$ and $x = -4$

**c.** $x = 6$ and $x = -8$

**d.** $x = 1$ and $x = 8$

To find the roots of the quadratic function $f(x) = x^2 - 6x + 8$, we set $f(x) = 0$ and solve for $x$. This gives us the equation $x^2 - 6x + 8 = 0$. We can factor this quadratic equation. We are looking for two numbers that multiply to 8 and add up to -6. These numbers are -2 and -4. Therefore, the factored form of the equation is $(x-2)(x-4) = 0$. Setting each factor to zero, we get $x - 2 = 0$ or $x - 4 = 0$. Solving these equations, we find $x = 2$ and $x = 4$. These are the roots of the function.

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**41** [Challenging] • Target: Series and sigma notation • Cross-Lesson: 2.01 Functions: Composite, Inverse & Logarithmic

A complex number $z$ is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

**a.** $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$ ✓

**b.** $5 + i5$

**c.** $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

**d.** $5i$

The rectangular form of a complex number given in polar form $z = r(\cos(\theta) + i\sin(\theta))$ is $a + bi$, where $a = r\cos(\theta)$ and $b = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{3}$: $a = 5\cos(\frac{\pi}{3}) = 5 \times \frac{1}{2} = \frac{5}{2}$
$b = 5\sin(\frac{\pi}{3}) = 5 \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}$ Therefore, the rectangular form is $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$.

**42** [Challenging] • Target: Geometric series • Cross-Lesson: 4.01 Lines, Triangles & Polygons

Consider the quadratic equation $x^2 - 8x + 15 = 0$. What are the roots of this equation?

**a.** $x = 3$ **and** $x = 5$ ✓

**b.** $x = -3$ and $x = -5$

**c.** $x = 3$ and $x = -5$

**d.** $x = -3$ and $x = 5$

To find the roots of the quadratic equation $x^2 - 8x + 15 = 0$, we can factor the quadratic expression. We are looking for two numbers that multiply to 15 and add up to -8. These numbers are -3 and -5. Therefore, the factored form of the equation is $(x - 3)(x - 5) = 0$. Setting each factor to zero, we get $x - 3 = 0$ which gives $x = 3$, and $x - 5 = 0$ which gives $x = 5$. Thus, the roots are $x = 3$ and $x = 5$.

**43** [Challenging] • Target: Complex number arithmetic • Cross-Lesson: 5.01 Trigonometry

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

**a.** **5** ✓

**b.** 7

**c.** 25

**d.** 16

The complex number can be written as $z = 3 + 4i$. The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. In this case, $a = 3$ and $b = 4$. Therefore, the modulus is
$|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

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**44** [Challenging] • Target: Complex conjugates and division • Cross-Lesson: 6.01 Circles & Coordinate Geometry

Consider the complex number $z = 3 + 4i$. What is the modulus of $z$?

**a.** **5** ✓

**b.** 7

**c.** 12

**d.** 25

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

**45** [Challenging] • Target: Modulus of complex numbers • Cross-Lesson: 7.01 Solid Geometry & Similarity

A quadratic with roots $r_1 = 2$ and $r_2 = 5$ has equation:

**a.** $x^2 - 3x + 10 = 0$

**b.** $x^2 - 10x + 7 = 0$

**c.** $x^2 - 7x + 10 = 0$ ✓

**d.** $x^2 + 7x + 10 = 0$

Applying Quadratics & Complex Numbers principles, $(x - 2)(x - 5) = x^2 - 7x + 10$. Therefore, the verified correct option is $x^2 - 7x + 10 = 0$.

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